PULLBACK PRESERVES SHORT EXACT SEQUENCES WHOSE QUOTIENT IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two related lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Pullback of a flat module is flat and Tor vanishes over flat modules. Stacks Project, Modules on Sites, Lemmas lemma-pullback-flat (tag 09BI) and lemma-flat-tor-zero (tag 09B8). Supplies the two inputs that keep exactness under tensoring with the structure sheaf.

Source reference. Original lemma 05VD: sources/stacks-chow/tags.txt, source lines 7547–7547. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03EX: sources/stacks-chow/tags.txt, source lines 4472–4472. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ be a morphism of ringed topoi. Let $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ be a short exact sequence of $\mathcal{O}$-modules with $\mathcal{H}$ a flat $\mathcal{O}$-module. Then the sequence $0 \to f^*\mathcal{F} \to f^*\mathcal{G} \to f^*\mathcal{H} \to 0$ is exact as well.*

Proof. Since f^{-1} is exact we have the short exact sequence $0 \to f^{-1}\mathcal{F} \to f^{-1}\mathcal{G} \to f^{-1}\mathcal{H} \to 0$ of $f^{-1}\mathcal{O}$-modules. By Lemma 05VD the $f^{-1}\mathcal{O}$-module $f^{-1}\mathcal{H}$ is flat. By Lemma 03EX this implies that tensoring the sequence over $f^{-1}\mathcal{O}$ with $\mathcal{O}'$ the sequence remains exact. Since $f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}} \mathcal{O}'$ and similarly for $\mathcal{G}$ and $\mathcal{H}$ we conclude. □

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